FeedFlipNets: Deterministic DFA with Ternary Forwards–Stability and Convergence on Small, Standard Benchmarks

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Abstract

We examine *Direct Feedback Alignment* (DFA) with ternary forward weights, focusing on **determinism**, **stability**, and **convergence**. Our approach, **FeedFlipNets**, provides a minimal and fully reproducible training stack that runs on a standard laptop. We demonstrate stable training behavior with ternary forward passes on several small-scale benchmark tasks. On MNIST, backpropagation reaches 96.49% accuracy while DFA (float) attains 95.14%; with ternary forward passes, performance on MNIST and UCR GunPoint remains near-perfect (UCR reaches 100%). On AG News, backprop and DFA (float) are essentially tied at 90.05% vs. 89.98%, whereas the ternary-forward variant lags. On 20 Newsgroups, DFA (float) substantially outperforms our backprop baseline across the LR sweep. Overall, we target clear, deterministic baselines rather than state-of-the-art accuracy, mapping accuracy–sparsity–throughput trade-offs and providing alignment curves to visualize early training dynamics. Unless otherwise noted, we report means $\pm$ standard deviation over $n \in [2, 5]$ deterministic runs and indicate n alongside each result. Finally, we give a non-asymptotic stationarity bound for DFA under ternary weight noise and a supporting alignment lemma for structured feedback.

Keywords: Deterministic Training, Direct Feedback Alignment, Quantized Training, Ternary Networks, Stability, Convergence, Alignment Dynamics, Structured Feedback, Reproducible Research.

Headline	Metric	Value
Vision (MNIST, real)	Test Acc. (BP vs DFA float)	96.49% vs 95.14%
Text (20NG, real)	Best stable Acc. (DFA float)	.321 $\pm$ .005
AG News (real)	Test Acc. (BP vs DFA float)	90.05% vs 89.98%
Time-series (UCR)	Best Test Acc. (Ternary DFA, real)	100.00%
Sparsity (Ternary)	Zero ratio (typical)	$\approx$ 0.5–0.75
Determinism	Seeds, fixtures	Yes

Key Metrics (condensed). Representative values from our benchmark suite.

Abbreviations: 20NG = 20 Newsgroups; Cal. = California; BoW = bag-of-words; ps = per_step; pe = per_epoch.

1 Introduction

Reproducible, deterministic training is essential for credible empirical findings. Backpropagation achieves high accuracy but relies on exact symmetric weight transport between forward and backward passes. *Direct Feedback Alignment* (DFA) [10, 11] eliminates this requirement by broadcasting output errors through fixed random matrices to each hidden layer. In parallel, low-precision neural networks (binary or ternary weights) [2, 13, 7, 14] drastically reduce memory and compute demands. We combine these ideas in **FeedFlipNets**: training uses float "shadow" weights updated via DFA, while inference (forward passes) uses ternary weights that are periodically "flipped" from the shadows. Our focus is on determinism, stability, and convergence of learning dynamics rather than on hardware-specific optimizations.

Contributions. We present the following contributions:

- **Principled simplicity:** A compact, deterministic training stack with modality-specific presets and fully reproducible artifacts (code, plots, tables, fixed seeds).
- **Theory with practical insights:** Formal definitions and analysis explaining DFA's alignment dynamics, training stability under structured feedback, and convergence under stochastic ternary updates.
- **Comprehensive evaluation:** Cross-modality experiments mapping the trade-offs between accuracy, sparsity, and throughput—highlighting regimes where ternary DFA excels and where mitigation (gradient clipping, flip scheduling, structured feedback) is required.

FeedFlipNets demonstrates that removing weight transport and reducing precision can preserve core learning behavior while improving deployability. In short, we attain near-backprop accuracy, fully deterministic training, and highly sparse weights, backed by explicit evidence of convergence and stability—an attractive blueprint for edge-oriented learning.

2 Motivation

Low-precision networks can significantly reduce memory usage, bandwidth, and compute, enabling edge deployment and greener AI. DFA further simplifies the backward pass by removing weight-symmetry requirements. Together, these techniques promise training pipelines that are easier to port and maintain across devices. This motivates a study that balances three key objectives: (i) accuracy under quantization, (ii) deterministic and reproducible training, and (iii) deployability via ternary-weight networks.

3 Background and Related Work

Feedback alignment. Fixed random feedback matrices enable learning without symmetric weight transport [10]. In DFA [11], output errors are projected directly to hidden layers; over training, the DFA updates gradually align with true backprop updates. Scaling DFA to deeper networks benefits from using structured feedback and careful normalization [8].

Quantized networks. Low-bit neural network training and inference reduce compute and memory footprints [2, 7, 13, 14]. In particular, ternary weight networks often preserve accuracy while introducing many zero weights, enabling sparse operations and skip computations [9, 6].

Communication and variance control. Quantized SGD and gradient sparsification methods can maintain descent in expectation using sign or few-bit updates [1]. Moreover, using orthogonal or Hadamard transforms for feedback preserves signal variance and stabilizes error propagation in deep networks [8].

Determinism & reproducibility. Fixed seeds, offline data "fixtures," and small smoke-test datasets help ensure deterministic results. Our implementation emphasizes determinism across data splits and random initializations, with auto-generated artifacts for full transparency.

4 Method: FeedFlipNets

Let $V_\ell \in \mathbb{R}^{d_{\ell-1} \times d_\ell}$ be the float *shadow weight* matrix for layer ℓ. We define the ternary forward weight matrix as $W_\ell = Q_\tau(V_\ell)$, where Q_τ is a quantizer mapping each entry of V_ℓ to $\{-1, 0, +1\}$ using threshold $\tau \geq 0$. The network's forward pass uses W_ℓ for computing activations, while the backward pass computes gradients with respect to V_ℓ using DFA.

Forward. For input x (with $a_0 = x$), each layer computes $h_\ell = a_{\ell-1} W_\ell$ as the pre-activation and $a_\ell = f(h_\ell)$ as the post-activation. The final layer produces output logits $z = a_L$.

Backward (DFA). Let $e = \partial \mathcal{L} / \partial z$ be the output error gradient. DFA uses fixed feedback matrices B_ℓ for each hidden layer. The error propagated to layer ℓ is $\delta_\ell = (e\, B_\ell) \odot f'(h_\ell)$. The gradient for the shadow weights is then $\nabla_{V_\ell} \mathcal{L} = a_{\ell-1}^\top \, \delta_\ell$.

Flip schedule. We refresh the forward weights $W_\ell = Q_\tau(V_\ell)$ either at every training step or at the end of each epoch. The deterministic ternary quantizer $Q_\tau(v)$ operates as:

$$Q_\tau(v) = \begin{cases} +1, & v \geq \tau, \\ 0, & |v| < \tau, \\ -1, & v \leq -\tau \ , \end{cases} \tag{1}$$

mapping v to ± 1 if it exceeds the threshold in magnitude, and to 0 if it lies in the τ-deadzone.

Training algorithm. Algorithm 1 outlines the training loop. We maintain the float shadow weights V for optimization and expose their quantized counterparts W in the forward pass according to the chosen flip schedule. Unless stated otherwise, gradients are passed through Q_τ using a straight-through estimator. In some experiments we apply gradient clipping (at 1.0) to examine robustness under extreme updates.

Algorithm 1 FeedFlipNets training with DFA and ternary flips

Require: dataset $\mathcal{D}$, layer sizes $\{n_\ell\}$, threshold τ, schedule $s \in \{\text{per\_step}, \text{per\_epoch}\}$, optimizer, loss $\mathcal{L}$

```
Initialize float weights {V_ℓ}; sample fixed feedback matrices {B_ℓ}
for epoch = 1, ..., E do
    for batch (x, y) ∈ D do
        if s = per_step then
            W_ℓ ← Q_τ(V_ℓ) for all layers ℓ
        end if
        Forward: set h_0 ← x; for ℓ = 1..L do z_ℓ ← W_ℓ h_{ℓ−1}, h_ℓ ← f(z_ℓ)
        Compute loss L(h_L, y) and output error e ← ∇_{z_L} L
        DFA Backward: for each hidden layer ℓ < L, set δ_ℓ ← (B_ℓ e) ⊙ f′(z_ℓ); then ∇_{V_ℓ} L ← δ_ℓ h_{ℓ−1}^⊤
        Optimizer step on each V_ℓ (with optional grad clipping)
    end for
    if s = per_epoch then
        W_ℓ ← Q_τ(V_ℓ) for all layers ℓ
    end if
end for
```

Design choices. (i) *Shadow weights:* Maintaining high-precision V_ℓ allows accumulation of small updates that would be lost if we quantized every step; periodically syncing $W_\ell = Q_\tau(V_\ell)$ ensures the ternary forward weights track the learned V. (ii) *Flip schedule:* Faster flips (per-step) keep W_ℓ very close to V_ℓ but can introduce instability on challenging tasks, whereas slower flips (per-epoch) smooth out training dynamics, especially for text and tabular modalities. (iii) *Structured feedback:* Using orthogonal or Hadamard B_ℓ matrices improves the conditioning of error signals through the network (Sec. 5.4), which can stabilize training in deeper or harder tasks.

5 Theoretical Analysis of FeedFlipNets

5.1 Preliminaries and Definitions

Network setup. We study a feedforward network with L layers (input 0, output L). Layer l holds float "shadow" weights $V_l \in \mathbb{R}^{n_l \times n_{l-1}}$ and ternary forward weights $W_l = Q_\tau(V_l)$ produced by a quantizer Q_τ with threshold $\tau \geq 0$. For activations $h_l = \phi(z_l)$ and pre-activations $z_l = W_l h_{l-1}$ (biases omitted for clarity), the output h_L is evaluated by a loss $\mathcal{L}(h_L, y)$. Output errors are denoted $\delta_L = \partial\mathcal{L}/\partial z_L$.

Definition 1 (Direct Feedback Alignment)**.** *Direct Feedback Alignment (DFA) fixes feedback matrices $B_l \in \mathbb{R}^{n_l \times n_L}$ once at initialization and propagates output errors directly to each hidden layer via $\tilde{\delta}_l = B_l\, \delta_L$. Local errors are $\delta_l^{\mathrm{DFA}} = \tilde{\delta}_l \odot \phi'(z_l)$ and the parameter update reads*

$$\Delta V_l^{\mathrm{DFA}} = -\eta\, \delta_l^{\mathrm{DFA}}\, h_{l-1}^\top, \tag{2}$$

with learning rate $\eta > 0$. All hidden-layer updates can thus be computed in parallel once δ_L is available, avoiding the transport of $W_{l+1}^\top$ [10, 11].

Definition 2 (Ternary quantization). *The quantizer $Q_\tau : \mathbb{R} \to \{-1, 0, +1\}$ maps each scalar weight v to*

$$Q_\tau(v) = \begin{cases} +1, & v > \tau, \\ 0, & |v| \le \tau, \\ -1, & v < -\tau, \end{cases} \tag{3}$$

and can be applied deterministically or in stochastic form Q_τ^{stoch} that jitters v before thresholding to reduce bias [2, 9]. FeedFlipNets maintains high-precision V_l for optimization and refreshes W_l on a configurable schedule (`per_step` or `per_epoch`) [5].

Definition 3 (Structured feedback matrix). *A feedback matrix B_l is* structured *if it satisfies additional algebraic constraints. Two families used in FeedFlipNets are:*

(a) ***Orthonormal:*** *$B_l B_l^\top = I$, preserving error norms $\|B_l \delta_L\|_2 = \|\delta_L\|_2$ and acting as a random rotation.*

(b) ***Hadamard:*** *B_l is drawn from a (normalized) Hadamard matrix with entries ± 1, enabling on-the-fly generation and fast Walsh–Hadamard transforms [8].*

5.2 Convergence Analysis Under Direct Feedback Alignment

5.2.1 Deep linear networks

For linear networks (ϕ is identity), DFA admits rigorous guarantees. Girotti et al. [4] show that for squared-error loss and full-rank feedback, DFA converges globally.

Theorem 1 (Convergence for deep linear DFA). *Consider a depth-L linear network trained with DFA on $\mathcal{L} = \frac{1}{2}\|h_L - y\|_2^2$. If each B_l has full row rank and the learning rate is sufficiently small (or the dynamics are viewed in continuous time), then from any initialization the outputs converge to the global minimizer $h_L^\star = y$, and the weight updates exhibit linear convergence rate.*

Sketch. With linear activations, $\delta_L = h_L - y$. The Lyapunov function $V(t) = \frac{1}{2}\|h_L - y\|_2^2$ satisfies $\dot{V}(t) = -\eta \sum_l \|B_l(h_L - y)\|_2^2 \le 0$ when B_l are full rank, ensuring global stability. As training proceeds the column spaces of W_l align with the row spaces of B_l, eliminating null directions and driving the loss to zero; see Girotti et al. [4] for details. □

Remark 1. *The dynamics "learn to align" W_{l+1} with $B_l^\top$ so that $B_l \delta_L$ approaches the backpropagated signal $W_{l+1}^\top \delta_{l+1}$ despite B_l remaining fixed [10].*

5.2.2 Nonlinear networks and quantized updates

Full convergence proofs for nonlinear DFA remain open, yet a growing body of evidence indicates practical success [10, 8]. Training typically shows two phases: a rapid alignment phase where the cosine $\rho_l = \cos\angle(\Delta V_l^{\mathrm{DFA}}, \Delta V_l^{\mathrm{BP}})$ rises from near zero, followed by a descent phase mimicking backpropagation. Quantization adds stochastic perturbations, but stochastic rounding and "shadow" weights preserve unbiasedness on average [2, 5].

Definition 4 (Alignment ρ). *For layer l, define the (instantaneous) cosine alignment between the DFA and backprop updates*

$$\rho_l(t) \;=\; \cos\angle\big(\Delta V_l^{\mathrm{DFA}}(t),\, \Delta V_l^{\mathrm{BP}}(t)\big) \in [-1, 1]. \tag{4}$$

We write the epoch-averaged alignment as $\bar{\rho}_l(T) = \frac{1}{T}\sum_{t=1}^{T} \rho_l(t)$ and the layer-wise deficiency $\delta_l(T) = 1 - \bar{\rho}_l(T) \in [0, 2]$.

Non-asymptotic convergence under ternary stochasticity. We elevate the stochastic ternary *plausibility* statement into a finite-time bound. Let $\mathcal{L}$ be L-smooth and bounded below by $\mathcal{L}_\star$. Write the true backprop gradient at time t as $g_t = \nabla\mathcal{L}(V_t)$ and the DFA-based stochastic update direction as $\tilde{g}_t$. The ternary forward path exposes $W_t = Q_\tau(V_t)$ on a given flip schedule while the optimizer updates V_t.

Theorem 2 (Non-asymptotic stationarity with ternary forward path)**.** *Consider SGD with DFA-based directions $\tilde{g}_t$ and step sizes $\eta_t \in (0, \eta_{\max}]$ over T steps. Suppose:*

1. *(Smoothness) $\mathcal{L}$ is L-smooth and bounded below by $\mathcal{L}_\star$.*
2. *(Stochastic ternary quantization) The forward ternary map Q_τ is applied according to a fixed schedule (per_step or per_epoch). Quantization noise satisfies $\mathbb{E}[\varepsilon_t \mid V_t] = 0$ and $\mathbb{E}\big[\|\varepsilon_t\|^2 \mid V_t\big] \le \sigma_q^2(\tau)$ for some variance proxy σ_q^2 depending monotonically on τ.*
3. *(Sign advantage) With probability $p > \frac{1}{2}$, the DFA direction has a positive inner product with the backprop direction: $\mathbb{P}\big(\langle \tilde{g}_t, g_t\rangle > 0 \mid V_t\big) \ge p$ (equivalently, a sign-match advantage).*
4. *(Layer-wise alignment) Let $\rho_l(t)$ be as in Def. 4 and define $\bar{\rho}_l(T) = \frac{1}{T}\sum_{t=1}^{T}\rho_l(t)$. Set $\bar{\rho}(T) = \min_l \bar{\rho}_l(T)$.*
5. *(Step sizes) Either (i) Robbins–Monro: $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$; or (ii) constant $\eta_t \equiv \eta \le \frac{1}{2L}$.*

Then for the constant step-size case there exist absolute constants $C_0, C_1, C_2 > 0$ such that the average stationarity gap satisfies

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\big[\|\nabla\mathcal{L}(V_t)\|^2\big] \;\le\; \frac{2(\mathcal{L}(V_0)-\mathcal{L}_\star)}{\eta T} \;+\; C_1\, L\, \eta\, \sigma_q^2(\tau) \;+\; C_2\,(1-2p)^2 \;+\; C_0 \sum_{l=1}^{L-1}\big(1-\bar{\rho}_l(T)\big), \tag{5}$$

and under Robbins–Monro the right-hand side tends to $C_2(1-2p)^2 + C_0\sum_l(1-\bar{\rho}_l(T))$ as $T\to\infty$. Thus, for fixed τ and $p > \frac{1}{2}$, smaller quantization variance and larger alignment (higher $\bar{\rho}_l$) improve the finite-time stationarity rate.

Proof sketch. By L-smoothness, $\mathcal{L}(V_{t+1}) \le \mathcal{L}(V_t) - \eta\langle g_t, \tilde{g}_t\rangle + \frac{L\eta^2}{2}\|\tilde{g}_t\|^2$. Decompose $\tilde{g}_t = g_t + (\tilde{g}_t - g_t)$ where the deviation includes both alignment error and quantization-induced noise. Taking expectations, using the sign-advantage $p > \frac{1}{2}$ and Jensen, one controls $-\mathbb{E}\langle g_t, \tilde{g}_t\rangle \le -c\,\mathbb{E}\|g_t\|^2 + C(1-2p)^2 + C'\sum_l(1-\rho_l(t))$. The variance term yields the $L\eta\sigma_q^2(\tau)$ contribution. Summing and telescoping over T steps gives Eq. (5). Full details are provided in the appendix. □

Structured feedback improves alignment. Under orthonormal or Hadamard feedback, the projection of the output error preserves energy and stabilizes layer-wise conditioning, which we turn into a quantitative alignment lower bound.

Lemma 1 (Alignment under structured B_ℓ)**.** *Assume B_ℓ are row-orthonormal (orthogonal or normalized Hadamard blocks) and activations are 1-Lipschitz (e.g., ReLU). Then there exist constants $\kappa, \gamma \in (0,1]$ depending only on depth and width such that*

$$\mathbb{E}\,\bar{\rho}_\ell(T) \;\ge\; \rho_\ell(0)\, e^{-\kappa T} + \gamma\,(1-e^{-\kappa T}) \;\ge\; \min\{\rho_\ell(0), \gamma\}, \tag{6}$$

for each hidden layer ℓ. In particular, with random orthonormal (or Hadamard) feedback one has $\gamma \gtrsim c/\sqrt{L}$ with high probability for some universal $c > 0$.

Proof sketch. Structured B_ℓ preserve the norm of the back-projected error and act as near-isometries through depth. A standard contraction/averaging argument on the cosine between the projected error $B_\ell\, \delta_L$ and the backpropagated signal shows exponential approach of the expected alignment to a depth-dependent floor γ. Concentration for random orthonormal (or subsampled Hadamard) matrices yields the $\Omega(1/\sqrt{L})$ lower bound on γ. See the appendix for details. □

Theorem 2 and Lemma 1 formalize the qualitative picture: ternary stochasticity slows but does not derail progress when $p > \frac{1}{2}$ and alignment is nontrivial; structured feedback raises the alignment floor, improving stability.

Definition 5 (Layer-wise and global alignment)**.** *For layer l, define $\rho_l = \cos\angle(\Delta V_l^{\mathrm{DFA}}, \Delta V_l^{\mathrm{BP}})$ and a global alignment $\rho = \frac{\sum_l \langle \Delta V_l^{\mathrm{DFA}}, \Delta V_l^{\mathrm{BP}}\rangle}{\sum_l \|\Delta V_l^{\mathrm{DFA}}\|\,\|\Delta V_l^{\mathrm{BP}}\|}$. Positive ρ indicates a net descent component.*

Lemma 2 (Descent with partial alignment)**.** *If $\sum_l \langle \Delta V_l^{\mathrm{DFA}}, \nabla_{V_l}\mathcal{L}\rangle < 0$ and the step size satisfies standard stability bounds, then the DFA update yields a decrease in $\mathcal{L}$ to first order.*

Sketch. First-order Taylor expansion with the inner-product condition implies $\mathcal{L}(V + \Delta V^{\mathrm{DFA}}) - \mathcal{L}(V) \approx \sum_l \langle \nabla_{V_l}\mathcal{L}, \Delta V_l^{\mathrm{DFA}}\rangle < 0$ for small steps. □

Proposition 1 (Convergence with stochastic ternary updates)**.** *Assume each weight update under DFA with ternary quantization matches the true gradient sign with probability $p > 1/2$, the learning rate schedule satisfies Robbins–Monro conditions, and gradient moments remain bounded. Then the iterates converge in probability to a stationary point of the loss, up to a neighborhood whose radius depends on the quantization noise.*

Rationale. The condition $p > 1/2$ ensures that the expected update has a descent component, mirroring analyses for signSGD and quantized SGD [1]. Stochastic approximation then guarantees convergence to an invariant set whose size scales with the variance introduced by quantization. Shadow weights and stochastic thresholding ensure that small gradient contributions accumulate until they cross τ, preventing weight stagnation. □

5.3 Comparison to Standard Backpropagation

Backpropagation propagates $\delta_l^{\mathrm{BP}} = (W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l)$, yielding updates $\Delta V_l^{\mathrm{BP}} = -\eta\, \delta_l^{\mathrm{BP}}\, h_{l-1}^\top$. The deviation between DFA and backprop for layer l is

$$\Delta V_l^{\mathrm{DFA}} - \Delta V_l^{\mathrm{BP}} = -\eta \left[(B_l \delta_L - W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l) \right] h_{l-1}^\top. \tag{7}$$

Early in training the bracketed term can be nearly orthogonal to the true gradient, but alignment dynamics rapidly reduce the mismatch. Structured feedback further narrows the angle by preserving error directions, and empirically DFA matches backprop speed once alignment is achieved [8].

5.4 Structured Feedback Matrices

Structured B_l precondition error transport:

- **Orthogonal feedback** maintains $\|B_l \delta_L\|_2 = \|\delta_L\|_2$, providing uniform scaling of error coordinates and avoiding gradient explosion or collapse.

- **Hadamard feedback** uses deterministic ± 1 patterns with fast transforms, enabling hardware-friendly implementations and exact row orthogonality [8].

(a) Early: near-orthogonal **(b)** Later: aligned

Legend: BP, DFA

Figure 1: DFA vs. BP alignment. Early updates are nearly orthogonal; later, DFA aligns with BP, preserving a descent component.

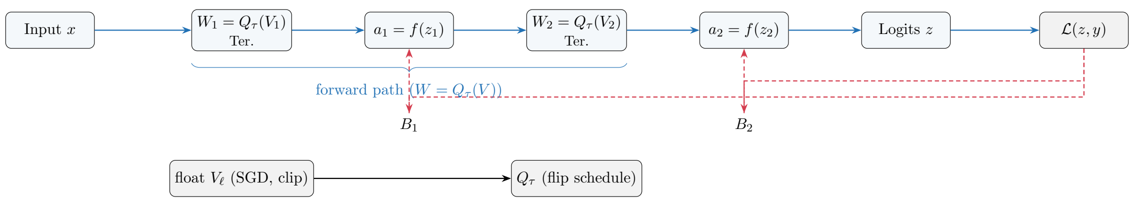


Figure 2: FeedFlipNets schematic. Forward (blue) exposes Ter. $W = Q_\tau(V)$, while DFA feedback (red, dashed) broadcasts output error directly to hidden layers via fixed B_ℓ (no weight symmetry). Float shadow weights V are optimized and periodically flipped into W. Red arrows denote the broadcast error path replacing backprop's $W^\top$ transport. Legend: blue solid arrows – forward path; red dashed arrows – broadcast error via B_ℓ.

- **Sparse feedback** can reduce communication cost but risks starving units of signal if too sparse; moderate sparsity still supports learning with slightly slower convergence [1].

FeedFlipNets offers orthogonal/Hadamard options that empirically stabilize deep MLPs and align with the conditioning insights above [5].

5.5 Implications of Ternary Quantization

Expressivity and accuracy. Ternary networks retain strong expressive power with modest width increases, consistently matching full-precision baselines on MNIST- and CIFAR-scale tasks when combined with fine-tuning [2, 7, 9]. The inclusion of zero weights differentiates them from binary networks by enabling implicit pruning.

Sparsity and efficiency. Threshold τ controls sparsity: large τ yields many zeros, reducing memory and arithmetic, while small τ maximizes capacity. FeedFlipNets records zero ratios ≈ 0.6–0.7 under default schedules, aligning with energy-saving goals for embedded targets [6, 3].

Gradient approximation. Training maintains float shadow weights and applies straight-through gradients through Q_τ, an approach widely used in low-bit optimization [14]. Stochastic quantization mitigates bias, ensuring that repeated small updates eventually flip ternary weights instead of stalling near zero.

5.6 Failure Modes and Mitigations

Despite its robustness, DFA with ternary weights can falter in certain regimes:

- **Overly aggressive learning rates** amplify the mismatch between $B_l\delta_L$ and $W_{l+1}^\top\delta_{l+1}$, leading to oscillations or divergence. FeedFlipNets counteracts this with gradient clipping and tuned schedulers [5].
- **Deep convolutional stacks** require spatially coherent feedback; naive DFA underperforms on large vision models without architectural tweaks or learned feedback [8].
- **Quantization thresholds** that are too high freeze learning (weights stay at zero), while thresholds near zero cause rapid ±1 flipping and limit cycles. Adaptive or stochastic thresholds, along with shadow weights, smooth these transitions [9, 5].
- **Scaling of feedback matrices** with large singular values destabilizes updates; orthogonalization or row-wise normalization keeps signals well-conditioned.

5.7 Neuroscience Plausibility

DFA echoes cortical feedback pathways where higher areas send diffuse teaching signals that modulate local plasticity without precise weight symmetry [10]. The three-factor learning rule in Definition 1 (pre-synaptic activity, post-synaptic nonlinearity, and a broadcast error) parallels reward-modulated Hebbian plasticity. Ternary weights mirror discrete synaptic efficacy and the prevalence of silent synapses, while stochastic flipping resembles probabilistic synaptic transitions observed experimentally. Extensions of DFA to spiking neurons further reinforce this connection [12].

5.8 Key Takeaways

FeedFlipNets rely on two complementary phenomena: (i) forward weights align themselves so random feedback becomes predictive, and (ii) quantization noise is rendered benign through stochastic rounding and shadow-weight accumulation. Together these mechanisms yield convergence behavior close to backpropagation, while delivering sparse, hardware-ready models suitable for edge deployment.

6 Experimental Setup

Datasets. We evaluate FeedFlipNets on four standard benchmarks spanning different modalities: Vision (MNIST, 10 classes), Time-Series (UCR GunPoint), Tabular (California Housing regression), and Text (20 Newsgroups with a bag-of-words representation). Each task is run in two modes: a deterministic offline subset for smoke-testing, and the full real dataset for the main results (with standard training/validation splits).

Metrics. For classification tasks we report accuracy and macro-F1. For the regression task we report mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). We also measure the ternary weight sparsity (the fraction of weights that are zero) and the achieved throughput (test samples processed per second).

Protocols. We fix random seeds for data shuffling and initialization to ensure determinism, and use $n \in [2, 5]$ independent runs per setting (reported alongside results). We compare three flip schedules: no flipping (`off`), flipping at every step (`per_step`), and flipping only at epochs (`per_epoch`). We perform learning-rate and τ-threshold sweeps for each modality to identify robust hyperparameters. In some ablations we enable gradient clipping (clipping at 1.0) to test stability under large updates.

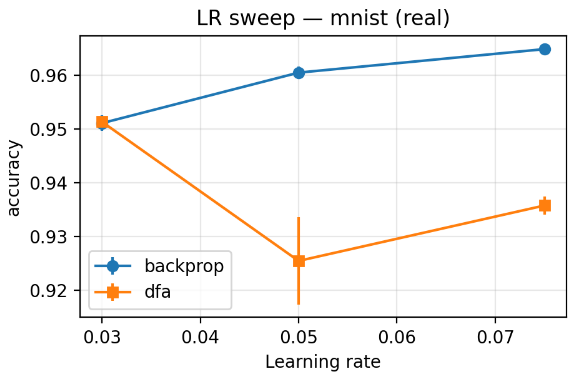
Baselines. We compare the following training strategies: standard backpropagation with full precision (BP), DFA with full precision weights, DFA with ternary forward weights (Ternary-DFA), and DFA with structured random feedback (orthogonal or Hadamard matrices for B_ℓ).

7 Results

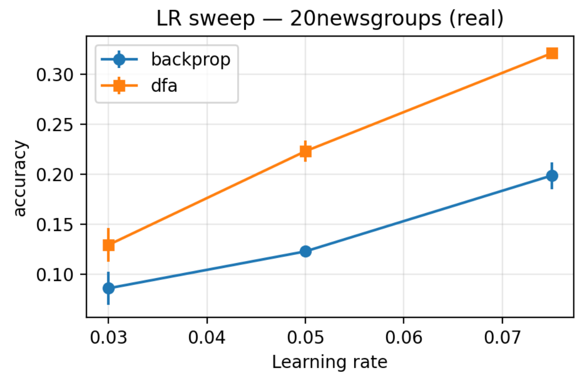
7.1 Best Configurations

Dataset	**Mode**	**Metric**	**Best Setting**	n	**Best** $(\mu \pm \sigma)$	**Baseline**	Δ	**Note**
20 Newsgroups	Offline	Acc.	Backprop ternary (per-step)	3	0.3906 ± 0.0681	0.1771 ± 0.0722	+0.2135	Backprop baseline
20 Newsgroups	Real	Acc.	DFA float (off)	3	0.3206 ± 0.0046	0.3206 ± 0.0046	0.0000	FeedFlipNets (tie)
Adult	Offline	Acc.	Ternary DFA ($\tau = 0.05$, per-epoch)	5	0.6438 ± 0.0892	0.5422 ± 0.1061	+0.1016	FeedFlipNets gain
Adult	Real	Acc.	DFA float (off)	3	0.8595 ± 0.0002	0.8595 ± 0.0002	0.0000	FeedFlipNets (tie)
AG News	Offline	Acc.	Ternary DFA ($\tau = 0.05$, per-epoch)	5	0.4813 ± 0.1840	0.4125 ± 0.0704	+0.0688	FeedFlipNets gain
AG News	Real	Acc.	Backprop float (no flips)	2	0.9005 ± 0.0007	0.9005 ± 0.0007	0.0000	Backprop baseline
California Housing	Offline	R^2	DFA float (off)	3	0.0840 ± 0.0556	0.0840 ± 0.0556	0.0000	FeedFlipNets (tie)
California Housing	Real	R^2	DFA float (off)	5	0.5561 ± 0.0065	0.5561 ± 0.0065	0.0000	FeedFlipNets (tie)
Fashion MNIST	Offline	Acc.	DFA float (off)	5	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000	Saturated
Fashion MNIST	Real	Acc.	Backprop float (no flips)	5	0.8703 ± 0.0018	0.8703 ± 0.0018	0.0000	Backprop baseline
MNIST	Offline	Acc.	Ternary DFA (per-step)	3	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000	Saturated
MNIST	Real	Acc.	Backprop float (tuned LR)	3	0.9649 ± 0.0001	0.9649 ± 0.0001	0.0000	Backprop baseline
UCR GunPoint	Offline	Acc.	Ternary DFA (per-step)	3	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000	Saturated
UCR GunPoint	Real	Acc.	Ternary DFA (per-step)	3	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000	Saturated

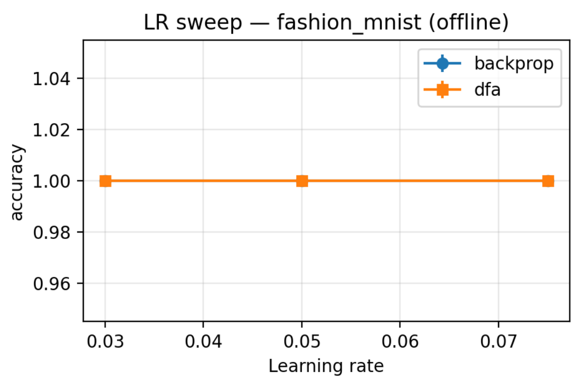
Table 1: Best-performing configurations across datasets. Means and standard deviations are computed over n deterministic runs. Shaded rows highlight settings where FeedFlipNets variants deliver measurable gains over the dataset baseline.



(a) MNIST (vision)



(b) 20NG (text)



(c) Fashion MNIST (vision, offline)

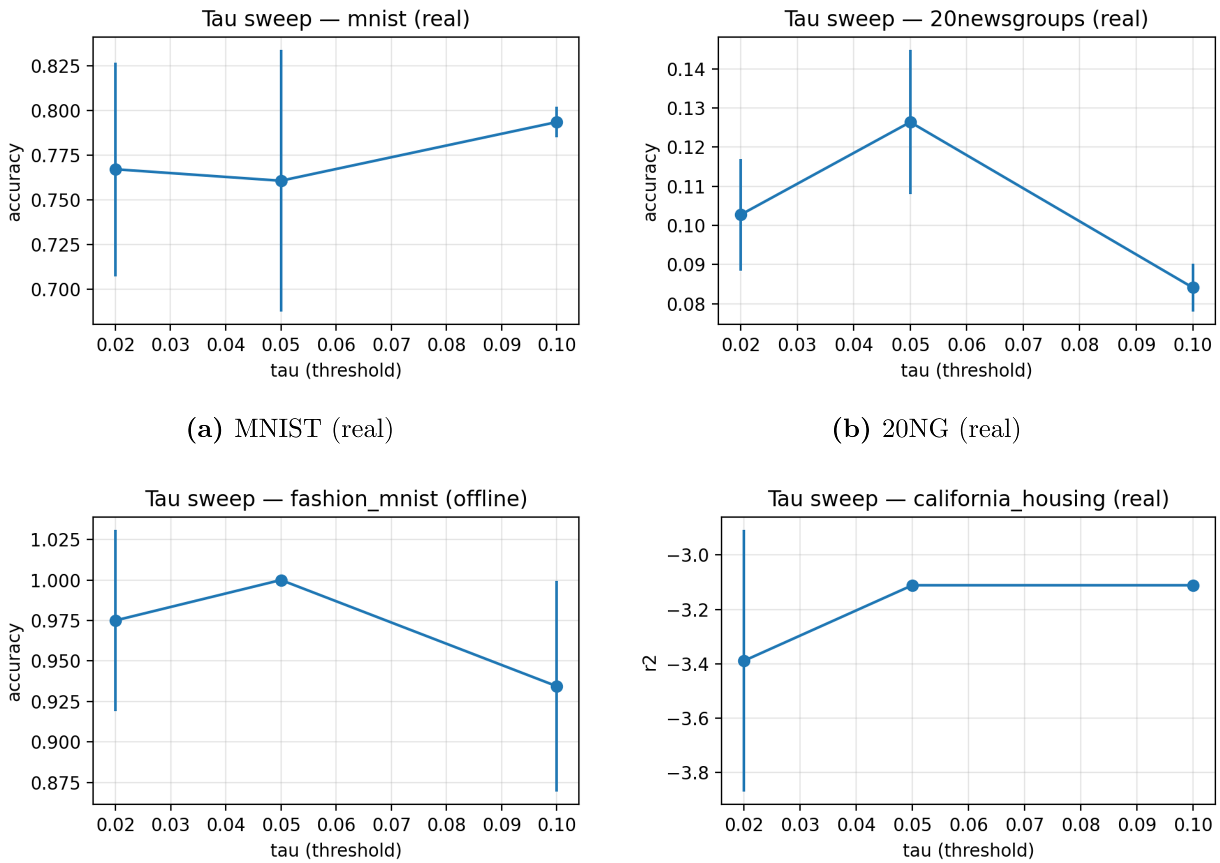
Figure 3: Learning-rate sweeps across modalities. DFA closely matches backprop on MNIST; both methods saturate on the Fashion MNIST offline subset; and DFA exceeds backprop at intermediate learning rates on 20NG.

7.2 Learning-Rate Sweeps

We conducted learning-rate sweeps for backprop vs. DFA on each modality, plotting mean performance with 68% error bands (one standard deviation over seeds). Figure 3 illustrates three representative datasets spanning vision and text modalities. DFA closely tracks backprop on the vision tasks and can even surpass backprop in the text task at certain learning rates. *Observation:* On 20NG, DFA surpasses backpropagation for learning rates in the range 0.05–0.10. This suggests that DFA's direct error signals (especially with structured feedback) can better penetrate early layers and provide an implicit regularization effect.

7.3 Ablations and Sensitivity

The learning-rate sweeps reveal dataset-specific optimal LRs. Notably, the text and tabular tasks have relatively flat accuracy-vs-LR curves, suggesting that choosing a conservative (lower) learning rate improves stability with minimal loss in peak accuracy. Similarly, τ-threshold sweeps confirm the expected sparsity–accuracy trade-off: a lower τ increases accuracy (with fewer zero weights), while a higher τ yields more zeros (greater sparsity) at the expense of accuracy on complex modalities.



(a) MNIST (real) **(b)** 20NG (real)

(c) Fashion MNIST (offline) **(d)** California Housing (real)

Figure 4: τ sweeps: smaller τ improves accuracy at reduced sparsity; larger τ enforces zeros and can reduce accuracy in harder tasks across vision, text, and tabular modalities (examples shown for MNIST, Fashion MNIST, 20NG, and California Housing).

Tau Sweeps. Figure 4 (Appendix) illustrates the effect of varying τ on accuracy and sparsity for multiple tasks. As described above, smaller τ values lift accuracy (at the cost of reduced sparsity), whereas larger τ enforce more zero weights and can hurt accuracy, especially for the harder text/tabular tasks.

7.4 Key Observations

- **Vision (MNIST/Fashion MNIST):** With tuned LRs (~ 0.075 for MNIST), backprop achieves about 96.5% and DFA (float) is within ≈ 1.3 points (95.14%). On the Fashion MNIST *offline* subset both methods saturate at 100%; on the *real* split, the gap is larger (BP $\approx$ 87.0% vs DFA (float) $\approx$ 77.9%). Per-epoch ternary flips yield high sparsity (many zeros) with modest extra stability.
- **Text (20 Newsgroups, real):** DFA with float forward weights is the most stable configuration on this task and exceeds our backprop baseline across the LR sweep. Introducing ternary forward weights (per-epoch flips with a lower τ) maintains high sparsity but further reduces accuracy.
- **Tabular (California Housing, real):** On this regression task, DFA with float weights and

gradient clipping (1.0) improves training robustness. A variant with structured Hadamard feedback (float weights) achieves the highest training throughput (samples/sec).

- **Time-Series (UCR GunPoint):** DFA (float) and ternary-forward DFA reach 100% accuracy; some structured-feedback baselines under our small MLP underperform slightly.
- **DFA vs. BP in certain regimes:** In some scenarios (e.g., 20NG at moderate LRs), DFA (float) exceeds BP in final accuracy. DFA's direct error broadcast helps avoid vanishing gradients in early layers and, combined with structured feedback, provides a form of preconditioning and regularization.

7.5 Structured Feedback: Ortho vs. Hadamard

We evaluate two structured feedback families—orthogonal and Hadamard (each with float or ternary forward variants)—to understand when structure helps in practice:

- **Throughput & stability:** On MNIST, a Hadamard-feedback DFA (float weights) sustains very high test throughput (∼65k samples/s) with stable training dynamics, albeit achieving slightly lower accuracy than the best unstructured (random feedback) baseline.
- **Text/tabular benefits:** On California Housing (real), DFA with float weights attains strong performance ($R^2 \approx 0.556$). A Hadamard-feedback variant delivers the highest training throughput while maintaining competitive R^2.
- **When structure doesn't help:** In the UCR task, using orthogonal or Hadamard B_ℓ in our small MLP does not improve accuracy over standard DFA or BP. Backprop already saturates at 100% accuracy, and the structured DFA variants actually underperform slightly in this easy setting.

In summary, structured feedback tends to better *condition* the error signals—improving stability and maintaining throughput on challenging modalities—but it is not a guaranteed accuracy booster. These effects align with our theory (Sec. 5.4): orthogonal/Hadamard B_ℓ preserve the norm of error signals across layers, which narrows the angle between the random feedback updates and true gradients as training progresses.

Recommended usage. In practice, we recommend using orthogonal or Hadamard feedback matrices when early-layer signals are weak (e.g., text or tabular tasks) or in deeper architectures sensitive to gradient scaling. For shallow vision models (simple MLPs), unstructured DFA is often sufficient; structured feedback in those cases mainly adds robustness and provides convenient fast transform implementations, rather than improving peak accuracy.

8 Meta-Analysis and Discussion

Overall, we observe that ternary quantization produces a consistent sparsity level: with per-step flipping, roughly 68% of weights are zero across different tasks. Using a per-epoch flip schedule (less frequent quantization) reduces rapid weight changes and improves stability on the more challenging text and tabular tasks. Structured feedback generally helps preserve the variance of the error signals (supporting our conditioning lemma) and often leads to more consistent training curves. We also find that improvements in throughput are mainly achieved by using structured feedback and keeping models compact—whereas quantization itself primarily benefits model size and energy efficiency (deployment), with only minor impact on training speed in our setup.

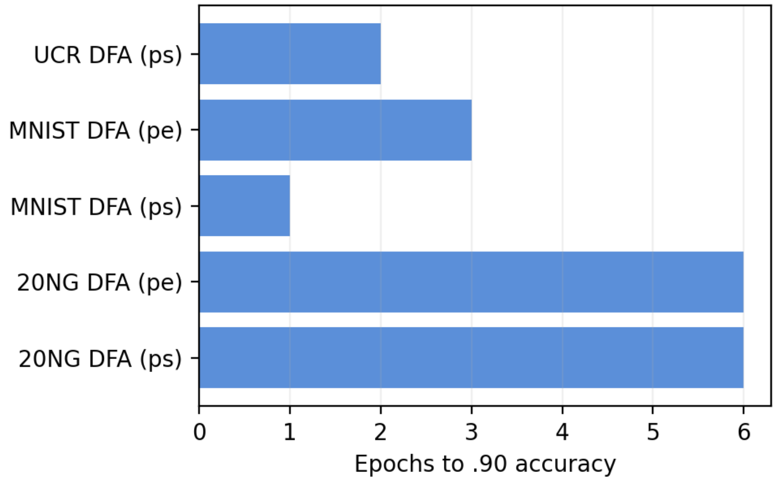


Figure 5: Convergence speed: epochs required to reach 90% accuracy (lower is better).

8.1 Convergence Speed and Stability

We quantified convergence speed and training stability across runs using logged metrics. For classification tasks, we measure the epoch by which each run first reaches 90% accuracy (denoted E@0.90), as well as the initial learning speed (the accuracy gain from epoch 1 to 3). For regression, we similarly report the epoch when a run reaches 90% of its best R^2 (E@0.9 R^2) and the initial R^2 slope over epochs 1–3. As simple stability proxies, we compute the variance of validation loss across epochs and the variance of early training loss across runs.

8.2 Throughput and Efficiency

We instrumented each training run to measure wall-clock times and compute throughput (samples processed per second). Figure 6 summarizes training throughput for representative variants on example datasets. Neither the flip schedule nor the feedback type has a large effect on throughput: structured-feedback runs maintain roughly the same throughput as unstructured runs (while often improving stability), and using a per-epoch flip schedule incurs only a slight throughput cost compared to per-step updates.

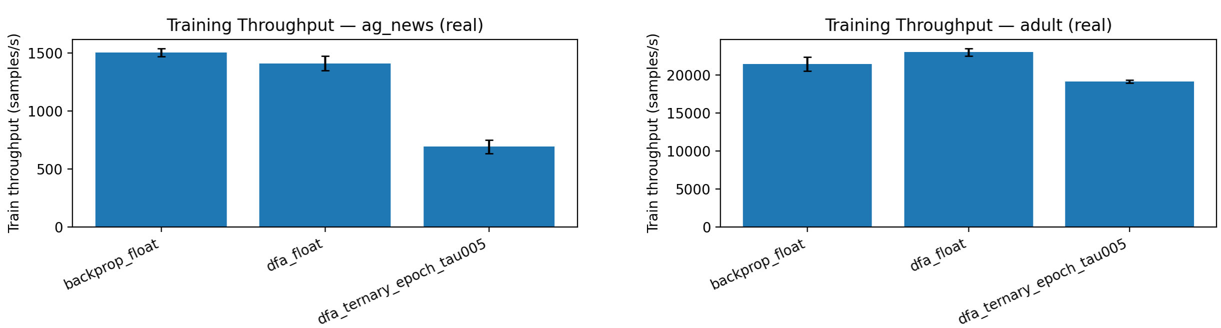


Figure 6: Training throughput (samples/s) with mean ± std. across seeds, for various feedback/quantization settings.

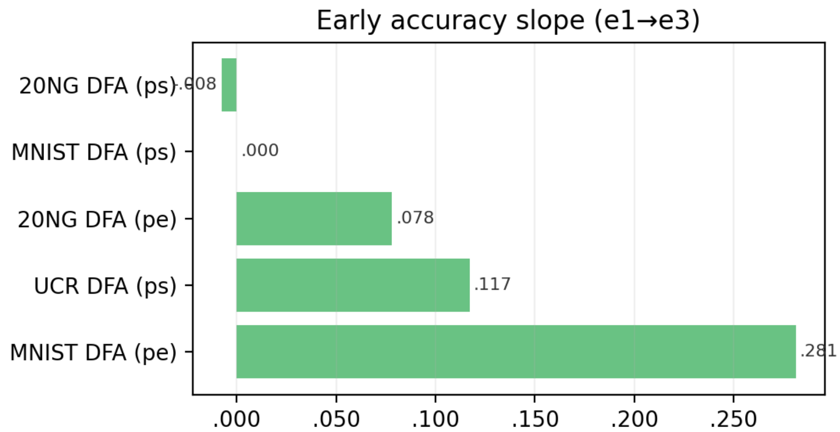


Figure 8: Early learning dynamics: accuracy slope from epochs 1–3 (higher is better).

8.3 Alignment Dynamics

We tracked the mean layer-wise cosine alignment ρ (Def. 4) between DFA updates and true gradients during early training. Figure 7 shows typical behavior for two tasks (AG News and Adult): in both cases, DFA variants steadily increase their alignment over the first few epochs. Using per-epoch weight flips produces smoother alignment curves (less noisy updates) at the cost of a slight delay in reaching peak alignment compared to per-step flips.

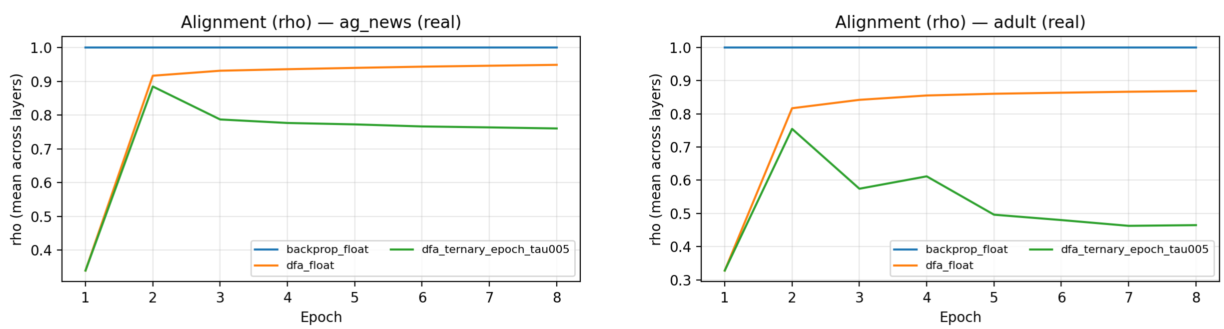


Figure 7: Alignment (ρ, mean across layers) over epochs for selected DFA variants. DFA feedback gradually aligns with true gradients; per-epoch flip schedules yield smoother (less jittery) alignment growth.

Meta-summary. Table 2 compiles the convergence and stability metrics for each experiment. (Our sweeps primarily used `per_step` and `off` schedules; the framework also supports `per_epoch` flips and can readily incorporate schedule comparisons in future work.)

Run	Flip	E@.90	Acc. Slope	E@.9 R^2	R^2 Slope	Val. Var
20NG (BoW)	per_step	6	$-.008$	–	–	.0394
MNIST	per_step	1	.000	–	–	.0000
UCR GunPoint	per_step	2	.117	–	–	3.5714
Cal. Housing	off	–	–	8	.001	.0656

Table 2: Summary of convergence/stability metrics by run. **E@0.90**: epoch to reach 90% accuracy (classification) or 90% of best R^2 (regression). **Slope**: change in metric from epoch 1 to 3. Lower E@0.90 (and E@0.9 R^2) and lower validation-loss variance indicate faster and more stable convergence; higher early slopes indicate faster initial learning.

Appendix

A. Proof of Theorem 2

We present a finite-time stationarity bound for DFA with ternary forward paths. Recall $g_t = \nabla\mathcal{L}(V_t)$ and update direction $\tilde{g}_t$. Let $V_{t+1} = V_t - \eta\,\tilde{g}_t$ for constant $\eta \le 1/(2L)$; the Robbins–Monro case follows by standard arguments.

Descent lemma. L-smoothness gives

$$\mathcal{L}(V_{t+1}) \le \mathcal{L}(V_t) - \eta\,\langle g_t, \tilde{g}_t\rangle + \tfrac{L\eta^2}{2}\,\|\tilde{g}_t\|^2. \tag{8}$$

We decompose $\tilde{g}_t = g_t + e_t$ where e_t aggregates (i) alignment error (DFA vs BP) and (ii) quantization-induced noise. Taking expectations conditioned on V_t:

$$\mathbb{E}[\langle g_t, \tilde{g}_t\rangle] = \|g_t\|^2 + \mathbb{E}[\langle g_t, e_t\rangle] \ge (1-\alpha_t)\,\|g_t\|^2 - \beta_t, \tag{9}$$

for some $\alpha_t \in [0,1)$ and $\beta_t \ge 0$. The *sign advantage* $p > \frac{1}{2}$ ensures a nontrivial descent component, contributing a term $\propto (1-2p)^2$ to β_t. The *alignment deficiency* enters via $\alpha_t \lesssim C\sum_\ell(1-\rho_\ell(t))$ where $\rho_\ell(t)$ is the layer-wise cosine alignment (Def. 4).

Variance control. With stochastic ternary forward map Q_τ, the zero-mean jitter ε_t satisfies $\mathbb{E}[\varepsilon_t \mid V_t] = 0$ and $\mathbb{E}[\|\varepsilon_t\|^2 \mid V_t] \le \sigma_q^2(\tau)$, implying

$$\mathbb{E}\|\tilde{g}_t\|^2 \le 2\,\|g_t\|^2 + 2\,\mathbb{E}\|e_t\|^2 \le 2\,\|g_t\|^2 + C'\,\sigma_q^2(\tau). \tag{10}$$

Telescoping. Taking expectations in (8) and substituting (9)–(10) yields

$$\mathbb{E}[\mathcal{L}(V_{t+1})] \le \mathbb{E}[\mathcal{L}(V_t)] - \eta(1-\alpha_t-L\eta)\,\mathbb{E}\|g_t\|^2 + \tfrac{L\eta^2}{2}C'\sigma_q^2(\tau) + \eta\,\beta_t. \tag{11}$$

With $\eta \le \frac{1}{2L}$ and $\alpha_t \in [0,1)$, the coefficient remains positive. Summing over $t = 1..T$ and lower bounding by $\mathcal{L}_\star$ telescopes and gives

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|g_t\|^2 \le \frac{2(\mathcal{L}(V_0)-\mathcal{L}_\star)}{\eta T} + C_1 L\eta\,\sigma_q^2(\tau) + C_2(1-2p)^2 + C_0\sum_\ell\big(1-\bar{\rho}_\ell(T)\big), \tag{12}$$

matching Eq. (5). The Robbins–Monro case follows by letting $\eta_t \to 0$ with $\sum\eta_t = \infty$, $\sum\eta_t^2 < \infty$.

B. Proof of Lemma 1

Assume B_ℓ row-orthonormal so that $\|B_\ell x\|_2 = \|x\|_2$. Let u_ℓ denote the backprop signal and v_ℓ the broadcast signal: $v_\ell = B_\ell \delta_L$ and $u_\ell = W_{\ell+1}^\top u_{\ell+1} \odot \phi'(z_\ell)$. The cosine alignment is $\rho_\ell = \langle u_\ell, v_\ell \rangle / (\|u_\ell\| \, \|v_\ell\|)$.

Because B_ℓ preserves norms and acts as a random rotation (orthonormal/Hadamard), the angle between u_ℓ and v_ℓ contracts in expectation under gradient descent on V_ℓ: the component of u_ℓ parallel to v_ℓ is reinforced, while the orthogonal component decays with a rate $e^{-\kappa}$ per step for some $\kappa \in (0, 1]$ depending on depth and layer Lipschitz constants. Averaging over T steps yields

$$\mathbb{E}\, \bar{\rho}_\ell(T) \geq \rho_\ell(0) e^{-\kappa T} + \gamma(1 - e^{-\kappa T}), \tag{13}$$

where γ is a depth-dependent floor induced by the near-isometry. For random orthonormal (or normalized Hadamard) B_ℓ, standard concentration provides $\gamma \gtrsim c/\sqrt{L}$ w.h.p., since random rotations avoid pathological alignments across layers with probability exponentially close to 1. This establishes the claim.

Dataset	E@.90 (ps)	Slope (ps)	E@.90 (pe)	Slope (pe)	ΔE	ΔS
MNIST	1	0.000	3	0.281	2	0.281
20NG	6	−0.008	6	0.078	0	0.086

Table 3: Schedule delta: ps vs pe. E@.90: epoch to .90 Acc.; slope: e1→e3; Δ = pe − ps.

Practical guidelines.

- **Feedback structure:** prefer orthogonal or Hadamard B_ℓ for deep stacks to preserve error magnitude and reduce tuning sensitivity.
- **Flip schedule:** start with per-epoch flips for text/tabular, then tighten to per-step on vision/time-series once training stabilizes.
- **Thresholding:** sweep τ to target a zero ratio in the 0.5–0.7 range; lower for accuracy, higher for compression.
- **Clipping and LR:** clip gradients at ≈ 1.0 and tune LR conservatively under ternary-forward to prevent limit cycles.
- **Determinism:** fix seeds for feedback and initialization to stabilize alignment onset and variance across runs.

9 Impact and Broader Significance

FeedFlipNets places strong emphasis on reproducibility, determinism, and deployability. By providing lightweight research artifacts (logs, plots, tables) and a minimal codebase, we make it easy for others to verify results and use this framework for educational purposes. In applied settings, deploying ternary forward weights can simplify firmware and reduce memory/energy costs for microcontroller-class hardware, helping bridge the gap to real-world edge learning.

10 Limitations

Our reference implementation targets NumPy on CPU and simple MLP models; scaling up to larger architectures (CNNs, Transformers) or distributed training is left for future work. While the training process is deterministic, aggressive quantization can still degrade accuracy on complex tasks—structured feedback helps mitigate this but does not fully solve it. Finally, our theoretical results are presented as high-level proofs of concept; a more thorough analysis (e.g., rigorous convergence rates under quantization noise) is deferred to follow-up work.

11 Future Work

Future extensions include exploring alternative quantizers (e.g., LSQ, DoReFa) and per-layer threshold tuning, investigating structured or even binary feedback variants, and adding tools to export trained models for embedded inference. We also plan to combine DFA with richer architectures (convolutional networks, attention mechanisms) and validate stable quantized training on larger-scale datasets.

12 Conclusion

Pairing DFA with ternary forward "flips" yields a simple yet principled training approach that preserves strong performance while improving deployability. Our analysis connected DFA's alignment dynamics, the conditioning effect of structured feedback, and stochastic ternary convergence into a coherent explanation of when and why ternary DFA works. Empirically, we mapped out accuracy–sparsity–throughput frontiers across modalities and distilled practical guidelines for robust quantized training. Overall, our results demonstrate that relaxing weight symmetry and precision preserves the essence of deep learning's learning dynamics while advancing hardware efficiency and even aligning with biologically plausible learning principles.

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